

SOUHAYLA ELMEFTAHI

DATA ENGINEERING STUDENT (GRADUATING IN JULY 2025)



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SUMMARY

Final-year data engineering student with strong knowledge in SQL, Python, data engineering tools, and cloud technologies (Azure). Eager to contribute to innovative projects while learning from industry experts during an end-of-year internship

PROFESSIONAL EXPERIENCE

Internship Data Enginner, Brocolidata

Jul 2024 - Sep 2024

- **Problem:** The start-up required a scalable and efficient data platform to handle large datasets in the cloud.
- **Solution:** Built a data platform on GCP, created a data lake with BigQuery, and automated workflows with Cloud Scheduler and FaaS.Impact: Improved data governance and reduced workflow management time.
- **Tools:** GCP, Docker, DuckDB, BigQuery, Cloud Scheduler, FaaS, Looker Studio,dbt.

Internship Data Science, 3D SMART FACTORY

Jul 2024 - Aug 2024

- **Problem:**Manual recruitment was slow and inefficient in evaluating candidates.
- **Solution:** Developed an NLP-based system that matched CVs to job descriptions, generated personalized questions, and provided real-time feedback with adaptive testing.
- **Tools:** Llama 3.1, NLP techniques, text similarity, semantic analysis,langchain.

Internship Machine Learning Engineer , IT department at ENSAH

Jun 2023- Aug 2023

- **Problem:** Manually analyzing and categorizing large volumes of research articles was inefficient and time-consuming.
- **Solution:** Scraped data from ResearchGate using Selenium and applied neural networks and unsupervised learning for clustering.Impact: Improved processing efficiency and article categorization accuracy.
- **Tools :** Selenium, Neural Networks, Pandas, NumPy, Matplotlib, Unsupervised Learning

PROJECTS

Building a Comprehensive Data Lakehouse Architecture for Optimized Data Management

- Architected a multi-layered system (Bronze, Silver, Gold) utilizing Minio for scalable storage, Apache Spark for data processing, Kafka for real-time data streaming, and various tools for advanced analytics and monitoring.
- **Tools :** Airflow, Python, Java, Minio, Apache Spark, Kafka, KRaft, AWS Lambda, Elasticsearch, Slack, Cassandra.

End-to-End Azure Cloud Migration and CI/CD Pipeline Implementation

- Connected a local PostgreSQL database to Azure, built a data pipeline using Azure Data Factory and Databricks, developed interactive reports with Power BI, and established automated testing and deployment workflows with Azure DevOps.
- **Tools :** SQL Server, Azure Data Factory, Data Lake, Synapse, Spark, Databricks, Power BI, Azure DevOps.

Developing Ensah_Aviation: An Open-Source Patent Analysis Tool for Sustainable Aviation Fuel

- There was a need for a free, easy-to-use tool to analyze patents, with a focus on aviation fuel
- Engineered a data pipeline with a medallion architecture using Spark for processing and transformations, automated workflows with Mage.ai, developed front-end features with HTML, CSS, and JavaScript, and designed interactive dashboards with Power BI.
- Improved research efficiency and contributing to technological monitoring
- **Tools:** Python, Selenium, Mage.ai, Snowflake, Spark, SVM, Power BI, HTML, CSS, JavaScript, Google Colab.

SKILLS

- **Programmation:** Python, SQL, PL/SQL, Java, Shell/Bash Scripting.
- **Cloud Platforms:** Azure, GCP (BigQuery, Cloud Scheduler, Looker Studio), AWS (S3, Lambda)
- **Data Engineering :** ETL, Hadoop Ecosystem, Spark, Kafka, Airflow, Selenium, Mage.ai, dbt, databricks
- **Data Science:** Machine Learning (Supervised learning/ Unsupervised learning), nlp, neural network.
- **Modeling/ Design:** UML, Merise.
- **Databases:** MySQL, SQL Server, Oracle, MongoDB, PostgreSQL.
- **Web Development & Tools:** HTML/CSS, Java EE, Spring, Docker, Git/GitHub, Linux, Windows
- **Leadership & Soft Skills :** Ownership, Curious, Think Big, Communication, Problem-Solving
- **Languages:** English, French, Arab.

EDUCATION

National School of Applied Sciences of Al- Hoceima	Sep 2020 - Jul 2025
• data engineering field	2022-2025
• Preparatory cycle: Sciences & Techniques for the Engineer	2020-2022
Faculty of Science and Technics of Al Hoceima	Sep 2019 - Jul 2020
• Mathematics, Computer Science, Physics	

CERTIFICATIONS

- **Azure For Data Engineering (5 Certificates)**
Issued by Microsoft (via Coursera)
- **Machine Learning - Python**
Supervised & Unsupervised Learning
Issued by DeepLearning.AI, Stanford University (via Coursera)
- **Big Data - Hadoop Ecosystem Masterclass**
Issued by Udemy
- **Data Engineering/Science/Analytics (12 Certificates)**
Issued by DataCamp

HOBBIES

- **Kickboxing Enthusiast.**
- **Debate Club Active Member and Data Club Member**
- **Avid Reader and Writer:** Enjoy reading a variety of genres and writing articles.